

University of Texas at Dallas
Department of Electrical Engineering
EEDG/CE 6306 - Application Specific Integrated Circuit Design
Homework #1

Submission :

- a) Written Report for Part 1, 2 and 3 (.pdf) (30pts)**
- b) Verilog design and testbench (30pts)**
- c) Synthesized Gate netlist and testbench (30pts)**
- d) Timing Report (10pts)**

Submit onto eLearning a zip file (team): <First name #1>_<Last name #1>_<First name #2>_<Last name #2>_HW01.zip

Submit onto eLearning a zip file (individual): <First name #1>_<Last name #1>_HW01.zip

If you have any question about the homework, you may contact the TA

Please include your partner's name in your report. There is one grade for each team. Only one team member is expected to submit the work onto eLearning. For this homework only, it is OK to make an individual submission if you have not found your team partner.

Purpose: Practice writing Verilog for combinational logic and testbenches. Practice running Synopsys Design Compiler (synthesis). Practice logic and gate simulations using modelsim.

Part 1 - Adder

Design a two-bit full adder in both behavioral and structural models. Also provide a testbench to verify its functionality.

- (1.1) Describe two-bit full adder in both behavioral and structural models in the report using Verilog..

(1.2) To verify functionality of two-bit full adder, test all possible inputs from your testbench.

- Preferred: Screenshot of your ModelSim waveform to verify the functionality of your two-bit full adder.
- But for this homework you can use any tools that you are familiar with, like Xilinx tools, edaplayground, etc.

(1.3) Include the verilog file and testbench in your ZIP file

Part 2 - Gatesim

Using Synopsys DC synthesize the two-bit full adder from Part 1 and show the critical path, timing report and waveform of gate-level netlist simulation.

(2.1) Follow the Design Compiler tutorial. Here are some specific guidelines for this assignment

- Skip step 6. Your adder is a fully combinational logic design for which timing constraints are not necessary.
- In step 7 and step 9, if you cannot see your design in “hier.” window, you can continue for now.
- In step 11, when simulating netlist, there will be lots of gates in “WORK” directory because we include “l8m.v” library. Be careful to find your own test bench.

(2.2) Your written report should include the schematic of the critical path of your adder.

(2.3) Include the generated gate netlist file and testbench in your ZIP file

(2.4) Include the timing report for the critical paths in your ZIP file

(2.5) Your written report should contain screenshots of the waveform of your modelsim gate netlist simulation exercising the critical paths of the adder.

Part 3 - Read the MSDAP paper

Read the MSDAP paper and write a summary of the ASIC function and the key benefits of the MSDAP algorithm. No more than 500 words.